

Week -2

Rayanki Himavarshini(2520030141)

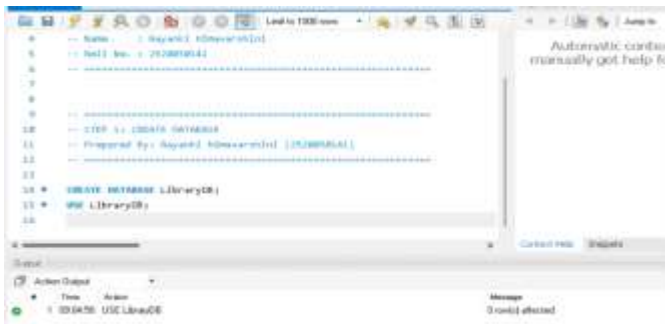
Week-2

The Librarian's Dashboard Scenario The library is now digital! The head librarian needs a report to see which members are active and which books are in the collection. You also need to handle a "Book Donation" process where a new book is added and assigned to a category at the exact same time—if one part fails, the whole process must stop to keep the catalog accurate.

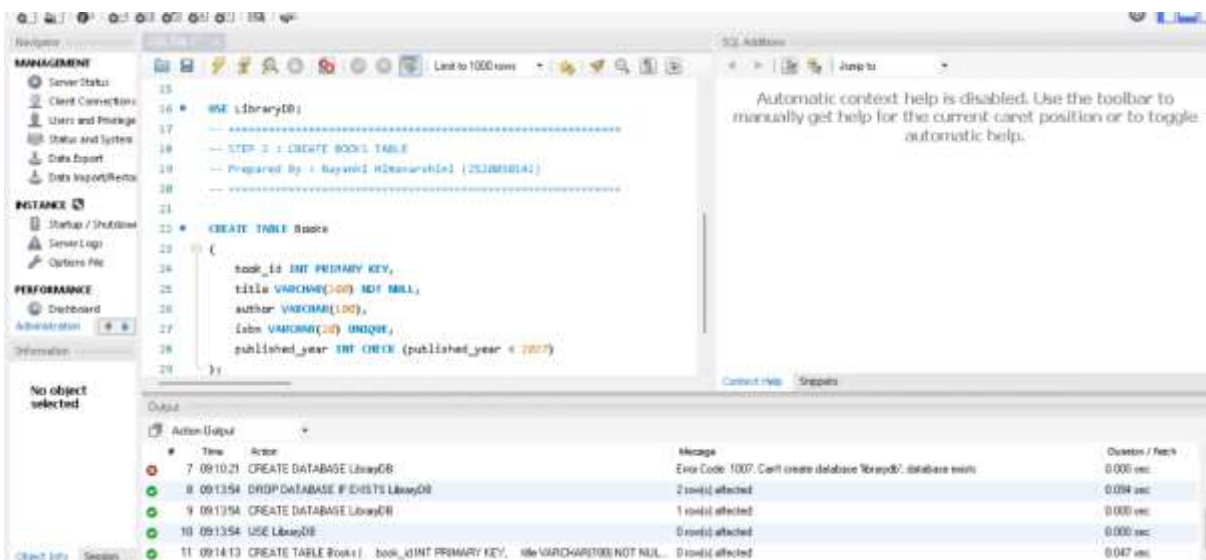
Task to be Done

1. The Catalog Search (JOINS): Create a third table called Loans that connects member_id to book_id. Write a query to show the Member Name and the Book Title they have currently borrowed.
2. Collection Stats (Aggregations): Write a query to find the total number of books published in each year. Hint: Use COUNT(book_id) and GROUP BY published_year.
3. The Donation Transaction (ACID): Use a Transaction (BEGIN...COMMIT) to add a new book to the Books table and simultaneously log the entry in a Donation_History table.
4. Search Speed (Indexing): Create an Index on the isbn column. Explain why searching by ISBN is now faster for the librarian than searching by Title. Skill Set Gained • Relational Logic: Connecting independent tables (Books + Members) through a Junction table (Loans). • Analytical SQL: Using GROUP BY to generate business insights. • Atomic Operations: Using Transactions to ensure multiple database changes succeed or fail as a single unit.

Step 1 — Create Database



Step 2 — Create Books Table (DDL + Constraints)



10 --
 11 -- STEP 3 : INSERT BOOK RECORDS
 12 -- Prepared By : Rayanki Himavanshi (2528030141)
 13 --
 14
 15 * INSERT INTO Books
 16 (book_id,title,author,isbn,published_year)
 17 VALUES
 18 (1,'The Great Book','P. Scott Fitzgerald','9780743273505',1925),
 19 (2,'To Kill a Mockingbird','Harper Lee','9780061120094',1960),
 20 (3,'1984','George Orwell','9780451524935',1949);
 21
 22 * SELECT * FROM Books;

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
21	09:16:21	USE LibraryDB	0 row(s) affected	0.000 sec
22	09:16:21	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL,	0 row(s) affected	0.021 sec
23	09:16:21	INSERT INTO Books (book_id,title,author,isbn,published_year) VALUES (1,'The Great Bo...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
24	09:16:21	SELECT * FROM Books LIMIT 0,1000	3 row(s) returned	0.015 sec / 0.000 sec
25	09:16:24	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL,	Error Code: 1050. Table 'books' already exists	0.000 sec

27 isbn VARCHAR(20) UNSIGNED;
 28 published_year INT CHECK (published_year < 2027)
 29);
 30 --
 31 -- STEP 3 : INSERT BOOK RECORDS
 32 -- Prepared By : Rayanki Himavanshi (2528030141)

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Context Help Snippets

Books 2

book_id	title	author	isbn	published_year
1	The Great Book	P. Scott Fitzgerald	9780743273505	1925
2	To Kill a Mockingbird	Harper Lee	9780061120094	1960
3	1984	George Orwell	9780451524935	1949

Output

Action Output

#	Time	Action	Message	Duration / Fetch
27	09:16:47	CREATE DATABASE LibraryDB	1 row(s) affected	0.015 sec
28	09:16:47	USE LibraryDB	0 row(s) affected	0.000 sec
29	09:16:47	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL,	0 row(s) affected	0.032 sec
30	09:16:47	INSERT INTO Books (book_id,title,author,isbn,published_year) VALUES (1,'The Great Bo...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
31	09:16:47	SELECT * FROM Books LIMIT 0,1000	3 row(s) returned	0.000 sec / 0.000 sec

SQL Editor

Limit to 1000 rows

Automatic context help is disabled. L manually get help for the current caret automatic help.

```

41
42 * SELECT * FROM Books;
43
44 -- STEP 4 : CREATE MEMBERS TABLE
45 -- Prepared By : Rayanki Himavanshini (2520030141)
46
47
48 * CREATE TABLE Members
49 (
50     member_id INT PRIMARY KEY,
51     full_name VARCHAR(100),
52     email VARCHAR(100) UNIQUE
53 );
54
55

```

Output

Action Output

#	Time	Action	Message
43	09:17:35	USE LibraryDB	0 row(s) affected
44	09:17:35	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NU...	0 row(s) affected
45	09:17:35	INSERT INTO Books (book_id,title,author,isbn,published_year) VALUES (1,'The Great Bo...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
46	09:17:35	SELECT * FROM Books LIMIT 0,1000	3 row(s) returned
47	09:17:35	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(10...	0 row(s) affected

SQL Editor

Filter Rows:

Result Grid

book_id	title	author	isbn	published_year
1	The Great Book	F. Scott Fitzgerald	9780743273565	1925
2	To Kill a Mockingbird	Harper Lee	9780061120084	1960
3	1984	George Orwell	9780451524935	1949
* HIDE	HIDE	HIDE	HIDE	HIDE

Books 6

Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message
52	09:18:10	USE LibraryDB	0 row(s) affected
53	09:18:10	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NU...	0 row(s) affected
54	09:18:10	INSERT INTO Books (book_id,title,author,isbn,published_year) VALUES (1,'The Great Bo...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
55	09:18:10	SELECT * FROM Books LIMIT 0,1000	3 row(s) returned
56	09:18:11	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(10...	0 row(s) affected

SQL File 1* x

Limit to 1000 rows

SQL Additions

Automatic context help is disabled. Use the manually get help for the current caret position or automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message
50	09:18:10	DROP DATABASE IF EXISTS LibraryDB	2 row(s) affected
51	09:18:10	CREATE DATABASE LibraryDB	1 row(s) affected
52	09:18:10	USE LibraryDB	0 row(s) affected

SQL File 1* x

Limit to 1000 rows

SQL Additions

Automatic context help is disabled. Use the manually get help for the current caret position or automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message
50	09:18:10	DROP DATABASE IF EXISTS LibraryDB	2 row(s) affected
51	09:18:10	CREATE DATABASE LibraryDB	1 row(s) affected
52	09:18:10	USE LibraryDB	0 row(s) affected
53	09:18:10	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, author VARCHAR(100), published_year INT)	0 row(s) affected
54	09:18:10	INSERT INTO Books (book_id,title,author,isbn,published_year)VALUES (1,'The Great Bo...'	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

SQL File 1

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

46
47 * CREATE TABLE Members
48 (
49     member_id INT PRIMARY KEY,
50     full_name VARCHAR(100),
51     email VARCHAR(100) UNIQUE
52 )

```

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05

Books 7 Members 8 Loans 9

Output

Action Output

#	Time	Action	Message	Duration / Fetch
65	09:20:09	INSERT INTO Members (member_id,full_name,email) VALUES (101,Rayanki Himavanshi)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
66	09:20:09	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
67	09:20:09	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT)	0 row(s) affected	0.062 sec
68	09:20:10	INSERT INTO Loans (loan_id,member_id,book_id,loan_date) VALUES (1,101,1,2025-01-05)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
69	09:20:10	SELECT * FROM Loans LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

135

136

Result Grid

book_id	title	author	isbn	published_year
1	The Great Book	F. Scott Fitzgerald	9780743273565	1925
2	To Kill a Mockingbird	Harper Lee	9780061120084	1960
3	1984	George Orwell	9780451524935	1949
NULL	NULL	NULL	NULL	NULL

Books 10

Output

Action Output

#	Time	Action	Message
66	09:20:09	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
67	09:20:09	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT)	0 row(s) affected
68	09:20:10	INSERT INTO Loans (loan_id,member_id,book_id,loan_date) VALUES (1,101,1,2025-01-05)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
69	09:20:10	SELECT * FROM Loans LIMIT 0, 1000	5 row(s) returned
70	09:20:55	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned

131

132

133

134

135

136

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```

131     donor_name VARCHAR(100),
132     donation_date DATE,
133
134     FOREIGN KEY(book_id) REFERENCES Books(book_id)
135 )
136

```

Member_Name	Book_Title
Rayanki Himavanshi	The Great Book
Rayanki Himavanshi	To Kill a Mockingbird
Emma Wilson	To Kill a Mockingbird
Emma Wilson	The Great Book
Michael Brown	1984

Books 16 Members 17 Loans 18 Result 19 Result 20

Output

Action Output

#	Time	Action	Message	Duration / Fetch
97	09:21:29	INSERT INTO Loans (loan_id,member_id,book_id,loan_date) VALUES (1,101,1,2025-01-05)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec
98	09:21:29	SELECT * FROM Loans LIMIT 0, 1000	5 row(s) returned	0.000 sec
99	09:21:29	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans INNER JOIN Members m ON m.member_id = loan_id JOIN Books b ON b.book_id = loan_id	5 row(s) returned	0.000 sec
100	09:21:29	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year	3 row(s) returned	0.000 sec
101	09:21:29	CREATE TABLE Donation_History (donation_id INT PRIMARY KEY, book_id INT, member_id INT, donation_date DATE)	0 row(s) affected	0.032 sec

134 FOREIGN KEY(book_id) REFERENCES Books(book_id)

135))

136

Result Grid

published_year	Total_Books
1925	1
1949	1
1960	1

Books 16 Members 17 Loans 18 Result 19 Result 20 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message
97	09:21:29	INSERT INTO Loans (loan_id,member_id,book_id,loan_date) VALUES (1,101,1,'2025-01-...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
98	09:21:29	SELECT * FROM Loans LIMIT 0,1000	5 row(s) returned
99	09:21:29	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN...	5 row(s) returned
100	09:21:29	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY pu...	3 row(s) returned
101	09:21:29	CREATE TABLE Donation_History (donation_id INT PRIMARY KEY, book_id INT, ...	0 row(s) affected

Result Grid

donation_id	book_id	donor_name	donation_date
1	4	Raj Kumar	2026-08-07

Donation_History 31 Books 32 Books 33 Members 34 Loans 35 Donate Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Pct
130	09:23:59	SELECT * FROM Books WHERE idno=9780451525342 LIMIT 0,1000	1 row(s) returned	0.000 sec / 0
139	09:23:59	SELECT * FROM Books LIMIT 0,1000	4 row(s) returned	0.000 sec / 0
140	09:23:59	SELECT * FROM Members LIMIT 0,1000	3 row(s) returned	0.000 sec / 0
141	09:23:59	SELECT * FROM Loans LIMIT 0,1000	5 row(s) returned	0.000 sec / 0
142	09:23:59	SELECT * FROM Donation_History LIMIT 0,1000	1 row(s) returned	0.000 sec / 0

Result Grid

donation_id	book_id	donor_name	donation_date
1	4	Raj Kumar	2026-08-07

Donation_History 42 Books 43 Books 44 Members 45 Loans 46 Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Pct
166	09:25:00	SELECT * FROM Members LIMIT 0,1000	3 row(s) returned	0.000 sec / 0.000 sec
167	09:25:00	SELECT * FROM Loans LIMIT 0,1000	5 row(s) returned	0.000 sec / 0.000 sec
168	09:25:00	SELECT * FROM Donation_History LIMIT 0,1000	1 row(s) returned	0.000 sec / 0.000 sec

Result Grid

book_id	title	author	isbn	published_year
1	Animal Farm	George Orwell	9780451526342	1945

Output

Action Output

#	Time	Action	Message	Duration / Fetch
166	09:25:00	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
167	09:25:00	SELECT * FROM Loans LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
168	09:25:00	SELECT * FROM Donation_History LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

SQL File

```

147 VALUES
148 (1,1,'B', 'Kumar',CURDATE());
149 COMMIT;
150 -- If any error occurs
151 ROLLBACK;
152

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

book_id	title	author	isbn	published_year
1	The Great Book	P. Scott Fitzgerald	9780743272565	1925
2	To Kill a Mockingbird	Harper Lee	9780061120084	1960
3	1984	George Orwell	9780451524935	1949
4	Animal Farm	George Orwell	9780451526342	1945

Output

Action Output

#	Time	Action	Message	Duration / Fetch
151	09:25:00	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
152	09:25:00	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT)	0 row(s) affected	0.031 sec
153	09:25:00	INSERT INTO Loans (loan_id,member_id,book_id,loan_date) VALUES (1,101,1,2025-01-10)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
154	09:25:00	SELECT * FROM Loans LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

File Edit View Query Database Server Tools Scripting Help

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System
- Data Export
- Data Import/Restore

INTAKE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Administration
- Information

No object selected

Output

Action Output

#	Time	Action	Message	Duration / Fetch
97	09:21:29	INSERT INTO Loans (loan_id,member_id,book_id,loan_date) VALUES (1,101,1,2025-01-10)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec
98	09:21:29	SELECT * FROM Loans LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
99	09:21:29	SELECT m.loan_id AS Member_Name, b.title AS Book_Title FROM Loans INNER JOIN Members m ON m.loan_id = b.loan_id	5 row(s) returned	0.000 sec / 0.000 sec
100	09:21:29	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year	2 row(s) returned	0.000 sec / 0.000 sec
101	09:21:29	CREATE TABLE Donation_History (donation_id INT PRIMARY KEY, book_id INT, member_id INT)	0 row(s) affected	0.032 sec
102	09:22:17	DROP DATABASE IF EXISTS LibraryDB	4 row(s) affected	0.018 sec
103	09:22:17	CREATE DATABASE LibraryDB	1 row(s) affected	0.016 sec
104	09:22:17	USE LibraryDB	0 row(s) affected	0.000 sec
105	09:22:17	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(255) NOT NULL)	0 row(s) affected	0.062 sec
106	09:22:17	INSERT INTO Books (book_id,title,author,isbn,published_year) VALUES (1,'The Great Book','P. Scott Fitzgerald','9780743272565','1925')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
107	09:22:17	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
108	09:22:17	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(255))	0 row(s) affected	0.032 sec
109	09:22:17	INSERT INTO Members (member_id,full_name) VALUES (101,'Rajankumar Harnavardhan')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
110	09:22:17	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
111	09:22:17	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT)	0 row(s) affected	0.047 sec
112	09:22:17	INSERT INTO Loans (loan_id,member_id,book_id,loan_date) VALUES (1,101,1,2025-01-10)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
113	09:22:17	SELECT * FROM Loans LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
114	09:22:17	SELECT m.loan_id AS Member_Name, b.title AS Book_Title FROM Loans INNER JOIN Members m ON m.loan_id = b.loan_id	5 row(s) returned	0.000 sec / 0.000 sec
115	09:22:17	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year	3 row(s) returned	0.000 sec / 0.000 sec
116	09:22:17	CREATE TABLE Donation_History (donation_id INT PRIMARY KEY, book_id INT, member_id INT)	0 row(s) affected	0.032 sec
117	09:22:59	DROP DATABASE IF EXISTS LibraryDB	4 row(s) affected	0.034 sec

SQL File 1

```
147 VALUES
148 (1,4,'Ra Kumar',CURDATE());
149 COMMIT;
150 -- If any error occurs
151 -- ROLLBACK;
152 --
```

Automatic context help is disabled. Use the tool manually get help for the current caret position or automatic help.

Result Grid

member_id	full_name	email
101	Rayenki Henuvarshi	2520030141@kth.edu.in
102	Emma Wilson	emma@gmail.com
103	Michael Brown	michael@gmail.com

Donation_History 42 Books 43 Books 44 Members 45 Loans 46

Output

Action Output

#	Time	Action	Message	Duration
97	09:21:29	INSERT INTO Loans (loan_id,member_id,book_id,loan_date) VALUES (1,101,1,2025-01-...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015
98	09:21:29	SELECT * FROM Loans LIMIT 0, 1000	5 row(s) returned	0.000
99	09:21:29	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOI...	5 row(s) returned	0.000
100	09:21:29	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY pu...	3 row(s) returned	0.000

Next Info Session